ISMS - Business Management System

Application Documentation

Version: 1.0.0
Platform: Cross-Platform (Web / Desktop / Android)
Release Type: Production-Ready Enterprise Application

1. OVERVIEW

ISMS (Inventory & Sales Management System) is a production-ready, cross-platform enterprise management application purpose-built for dairy farms, milk processing plants, and retail dairy outlets. It delivers an online-first, real-time reactive architecture that synchronizes business operations across every device in the organization instantly.

The application couples a modern React SPA frontend with a Supabase (PostgreSQL) real-time backend, wrapped into native desktop applications via Tauri v2 (Windows/Linux) and mobile apps via Capacitor v8 (Android). It is designed specifically to handle the unique operational challenges of the Kenyan dairy industry - including perishable inventory, M-Pesa digital payments, cold-chain logistics, and multi-branch distribution.

2. CORE PURPOSE

ISMS is engineered to unify the entire dairy business lifecycle into a single connected platform:

- Manage Point-of-Sale (POS) operations with full cart, discount, and tax handling.
- Track perishable dairy inventory with batch/expiry monitoring and spoilage logging.
- Run customer loyalty programs with tiering, wallet deposits, and credit management.
- Coordinate field operations including riders, delivery routes, and shift management.
- Deliver executive analytics with AI-assisted business intelligence reporting.
- Enforce strict access control via row-level security (RLS) and role-based permissions.

3. TECHNOLOGY STACK

- Frontend Framework : React 19 + TypeScript (Core SPA UI)
- Build Tool : Vite 6 (Fast bundling and dev server)
- Styling : Tailwind CSS v4 (Utility-first design)
- State Management : Zustand 5 (Modular reactive stores)
- Backend/Database : Supabase (PostgreSQL 15) + realtime
- Realtime Sync : Supabase Realtime (WebSocket)
- Desktop Wrapper : Tauri v2 (Rust) - Windows/Linux
- Mobile Wrapper : Capacitor v8 - Android
- AI Engine : Google Gemini via Edge Functions
- Charts : Recharts (Data visualization)

- PDF Export : jsPDF (Invoices and reports)
- QR Codes : qrcode (Receipt and product encoding)
- Icons : lucide-react (Modern iconography)
- Animations : motion (UI transitions)

4. FEATURE MODULES

The application is organized into 17 feature viewports navigated via a tab-based shell.

4.1 POS Checkout (POSView)

The heart of daily sales operations:
- Walk-in and Loyalty Transactions: Process sales against registered customer profiles or as anonymous walk-ins.
- Cart Management: Dynamic quantity adjustment, line-item percentage discounts, and VAT computation.
- Payment Flexibility: Cash, M-Pesa, Card, Bank Transfer, Mobile Wallet, and Credit/Debt settlement.
- Delivery Support: Mark orders as deliveries with rider assignment and delivery fee shortcuts (KSh 0/150/250).
- Credit Sale Handling: Atomic database processing for wallet deduction and debt creation.
- Real-Time Stock Deduction: Inventory updated at sale completion.

4.2 Inventory Management (InventoryView)

Full dairy stock pipeline control:
- Product Directory: Name, category, SKU, cost, sale price, unit, and minimum stock thresholds.
- Expiry & Batch Tracking: Perishable flags, expiry days, and shelf-life risk alerts.
- Wastage Ledger: Record spillage, souring, and processing losses.
- Reconciliation Logs: Manual audits, damages, and restock arrivals.
- Low-Stock Detection: Visual alerts below safety buffer.

4.3 Sales & Receipt History (SalesView)

Complete transaction audit trail:
- Transaction Logs: Detailed sale history with items and payments.
- PDF Export: Professional invoice PDFs via jsPDF.
- Refund Handling: Transaction reversals and credit adjustments.
- Item-Level Detail: Product names, prices, quantities, and discounts.

4.4 Customers & Loyalty (CustomersView)

Membership and financial profile management:
- Loyalty Tiering: Automatic Bronze, Silver, Gold classification.
- Wallet System: Client deposits and wallet-based checkout.
- Credit / Debt Manager: Pending balances and payment allocations.
- Loyalty Points: Award on purchases, redeem on future orders.
- Delivery Subscriptions: Recurring daily/weekly milk deliveries.

4.5 Executive Analytics (BusinessDashboard)

AI-enhanced decision support:
- KPI Deck: Gross sales, net profit, inventory valuation, loyalty.
- Financial Matrix: Revenue, COGS, delivery, wages, overhead.

- Payment Mix: M-Pesa vs Cash vs Card vs Bank breakdown.
- Sales Trajectory: Area charts of revenue, expenses, profit.
- Inventory Pipeline: Restock recommendations, top sellers.
- Branch Performance: Multi-branch revenue comparison.
- Staff Metrics: Active shifts, checkout velocity, task completion.
- AI Advisory (bi-analyze): Executive summaries, insights, risks, opportunities. Supports direct Q&A ("How can I improve profits?").

4.6 Workers & Shift Control (WorkersView)

Workforce management:
- Shift Recorder: Punch-in/out with cash reconciliation.
- Task Checklists: Duty assignments for staff profiles.
- Sales Auditing: Per-shift sales totals and metrics.

4.7 Permissions & RBAC (PermissionsView)

Granular access control:
- Visual Accordion Editor: Toggle role permissions.
- Granular Codes: pos.create_sale, bom.view, etc.
- Instant Sync: Push configurations to Supabase in real time.

4.8 AI Workspace Assistant (WorkspaceAssistantView)

Intelligent copilot ("Kim"):
- Chat Interface: Conversational AI via Gemini.
- Action Triggers: create_schedule, adjust_stock, create_checkout.

4.9 Business Management (BusinessManagementView)

Tenant and brand administration:
- Business Profiles: Name, logo, currency, country, timezone.
- Branch Setup: Physical locations and distribution hubs.
- Business Type: Predefined or custom types (Butchery, Logistics).

4.10 Production & BOM (ProductionBOMView)

Manufacturing oversight:
- Bill of Materials: Recipes and production inputs.
- Production Batches: Batch logs with quantity and status.
- Recipe Management: Products linked to production recipes.

4.11 Communication Center (CommunicationCenterView)

Business messaging hub:
- SMS Dispatch: Transactional updates via Twilio.
- Customer Communications: Targeted messaging.
- Notification Logs: Outbound communication tracking.

4.12 Customer Feedback (CustomerFeedbackView)

Voice-of-customer management:
- Review Collection: Ratings and comments.
- Sentiment Analysis: Positive/neutral/negative classification.
- Complaint Tracking: Open vs resolved SLA cases.

4.13 Notifications Center (NotificationsView)

Alert management:
- Push Notifications: Payment, delivery, shift alerts.
- Real-Time Alerts: System-wide event broadcasts.

4.14 Settings Dashboard (SettingsView)

System configuration:
- Thermal Printer Setup: ESC/POS hardware configuration.
- SMS Configuration: Twilio credentials and templates.
- Receipt Settings: Paper size, QR, barcode, tax breakdown.
- Auto Backups: Sync to Google Sheets.
- Device Permissions: Camera, geolocation, Bluetooth, storage.

4.15 Profile View (ProfileView)

User account management:
- Profile Editing: Name, avatar, contact details.
- Preferences: App appearance and notifications.

4.16 Home & Dashboard (HomeView, DashboardView)

Entry points and overview:
- Home View: Quick navigation and recent activity.
- Dashboard: High-level operational metrics.

5. USER ROLES & ACCESS CONTROL MATRIX

Owner/Admin:
Master clearance. Full control over settings, billing, financial profiles, and staff registries. Overrides all system flags.

Manager:
Operational management. Oversees sales, stock, staff scheduling, expenses, and analytics. Business views, inventory adjustments, POS sales, customer ledger updates, reports.

Cashier:
Daily point-of-sale operator. POS sales, shift open/close, customer registration, wallet deposits, cash verification.

Rider:
Logistics and deliveries. Order logs, delivery zones, route status completion, local chat.

All data access is enforced server-side via PostgreSQL Row-Level Security (RLS) policies tied to tenant business memberships.

6. ARCHITECTURE & DATA FLOW

Frontend Architecture - Modular Zustand Stores:
- appStore: Unified workspace facade.
- authStore: Onboarding, multi-tenancy, authentication, shifts.
- businessStore: Tenant businesses and branch routing.
- cartStore: POS cart manipulation and checkout.
- customerStore: Loyalty profiles, credit, wallets.
- inventoryStore: Catalog, stock arrivals, wastage.
- transactionStore: Transaction and debt-payment records.
- notificationStore: App-wide notification routing.
- extraModulesStore: Purchases, production, recipes, assets.
- uiStore: Global UI state and chat records.

Repository Layer:

services/repositories.ts abstracts all database reads/writes with
real-time subscription callbacks.

Realtime Data Flow:
1. User performs an action (e.g., POS checkout).
2. Frontend store calls the repository method.
3. Repository writes to Supabase (PostgreSQL).
4. Supabase Realtime broadcasts the change via WebSocket.
5. All connected devices update their local stores instantly.

AI Pipeline:
1. Dashboard collects local metrics (sales, expenses, inventory).
2. Frontend invokes the bi-analyze Edge Function with metrics.
3. Edge function loads AI settings for the tenant.
4. Calls AI provider (Gemini, HuggingFace, OpenRouter).
5. Parses structured JSON response and returns executive report.
6. Dashboard renders insights, risks, and action checklist.

7. DATABASE SCHEMA

Core Tables:
- businesses: Tenant definitions (name, currency, logo).
- branches: Physical locations and distribution hubs.
- users: Master credential directory.
- employees: Staff profiles with shifts and tasks.
- business_memberships: User-to-business role assignments.
- invitations: Staff invitation tokens.
- role_permissions: Role-to-privilege mappings.
- products: Catalog with pricing, stock, min-stock.
- customers: Profiles with tier, wallet, credit balance.
- transactions: Checkout registers (totals, tax, delivery).
- transaction_items: Line items with product snapshots.
- shifts: Punch-clock durations and cash reconciliation.
- inventory_adjustments: Wastage and restock records.
- expenses: Operational cost tracking.
- suppliers: Raw milk vendor profiles.
- notifications: Push notification registry.
- receipt_settings: Receipt layout and branding.
- sms_settings / ai_settings: Provider credentials.
- google_sheets_backup: Cloud spreadsheet backup sync.

SQL Views:
- low_stock_products: Products below safety thresholds.
- active_credit_debts: Customers with pending balances.
- expense_category_summary: Expenses grouped by category.
- product_sales_ranking: Products by velocity and revenue.
- business_statistics: Real-time dashboard KPIs.

8. GETTING STARTED

Prerequisites:
- Node.js v20+
- Rust Compiler & Cargo (Tauri desktop builds)
- Android Studio & SDK Tools (Capacitor mobile builds)

- Supabase CLI (backend management)

Environment Configuration:
Create a .env file in the project root:
VITE_SUPABASE_URL=your_supabase_project_url
VITE_SUPABASE_ANON_KEY=your_supabase_public_anon_key
VITE_AI_NAME=Kim
VITE_UNSPLASH_ACCESS_KEY=optional_unsplash_api_key
GEMINI_API_KEY=your_google_gemini_api_key

Installation:
npm install

Running Development Servers:
- Web (Vite): npm run dev -> http://localhost:3001
- Desktop (Tauri): npm run tauri:dev
- Android Emulator: npm run android

9. PRODUCTION BUILDS

Desktop (Tauri):
npm run tauri:build
- Windows: NSIS .exe installers + .msi packages.
- Linux: .AppImage + .deb packages.

Mobile (Capacitor):
./build-local.sh Compile unsigned release APK
./sign-apk.sh Align and sign the release APK
- Signed APK: release-builds/android/isms-release-signed.apk
- Install by sharing and tapping the APK on your phone.

10. BACKEND DEPLOYMENT (SUPABASE)

supabase link --project-ref your-project-ref-id
supabase db push
supabase secrets set GEMINI_API_KEY="your-gemini-key"
supabase functions deploy bi-analyze
supabase functions deploy chat
supabase functions deploy mpesa-callback

11. SECURITY & COMPLIANCE

- Row-Level Security (RLS): Every tenant's data is isolated behind PostgreSQL policies.
- Role-Based Access Control (RBAC): Granular permission codes gate each operation.
- Multi-Tenancy: Businesses and branches are fully isolated.
- Real-Time Intent: Online-first architecture for critical operations.
- Atomic Transactions: Credit sales use single database procedures to prevent inconsistent wallet/debt states.

12. CONCLUSION

ISMS is a modern, enterprise-grade dairy business management platform that combines real-time operations, deep analytics, AI-assisted

decision making, and robust multi-tenant security into a unified cross-platform experience. It helps Kenyan dairy businesses - from single retail outlets to multi-branch processing enterprises - operate more efficiently, track profitability, retain customers, and make smarter data-driven decisions.